Fiscal Year:	FY 2015	**Task Last Updated:**	FY 09/19/2014
PI Name:	Karma, Alain Ph.D.		
Project Title:	Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
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No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
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Flight Program:			
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Key Personnel Changes/Previous PI:			
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Performance Goal No.:			
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	EDITOR's NOTE (8/20/2014): Continuation of "Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys"--grant NNX11AC09G. New grant is to foster collaboration with ESA partners on the CETSOL and XRMON projects. Cellular and dendritic microstructures formed during industrial solidification processes (such as casting, welding, and brazing) determine the properties of commercial metallic alloys produced in numerous industries, with applications ranging from automobile engine parts to aircraft turbine blades. The production of innovative materials and the optimization of both properties and processing routes go through an accurate prediction and modeling of the dynamics of microstructures formation. From a fundamental standpoint, the formation of these structures is controlled by a subtle interplay between microscopic interfacial phenomena and macroscopic transport of heat and solute, which is strongly influenced by gravity-induced melt convection under terrestrial conditions. Therefore, microgravity directional Bridgman		

Task Description:	experiments provide a unique means to investigate microstructure development with negligible convection and to test the predictions of theoretical models in a purely diffusive growth regime. While most of the theoretical research to date has focused on the growth of single crystals, most commercially relevant metallic alloys are polycrystalline. The main aim of this research is to bring a quantum leap in our ability to model realistically dendritic alloy structures in polycrystalline materials where interactions between well-developed dendrites from the same grain and from grains of different crystal orientations both need to be considered. Interactions at grain boundaries are especially important because of their influence on dendrite fragmentation and the formation of solidification defects. The present project is an extension of the project NNX11AC09G, ending in 2014. The main achievement of this previous project is the development of a new multi-scale modeling approach to model dendritic solutal interactions at a millimeter scale while conserving the details of the grain structures (secondary, tertiary, and higher order side-branches). This "Dendritic Needle Network" (DNN) model fills the gap between the phase-field technique, relevant at a microscopic scale smaller than dendritic tip, and cellular-automaton approaches studying solidification processes at the scale of individual grains. This project consists mainly of using extensively the computational tools previously developed in order to investigate quantitatively mechanisms of dendritic growth in metallic alloys such as the competitive dynamics of side-branches, elimination processes of misoriented grains in directional solidification and the columnar-to-equiaxed transition. Simulations results will be compared to microgravity and ground based experiments in well-characterized aluminum based binary alloys, such as Al-Cu and Al-Si. Those studies will also set the stage for planning future microgravity Bridgman experiments on the International Space Station that use bi-crystalline samples to investigate both intragrain and intergrain dendrite interactions in a diffusive growth regime.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based superalloys used in turbine blades to light weight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process. These microstructures in turn control the integrity and mechanical properties of cast parts. This research is aimed at developing computational tools to predict reliably the formation of these dendritic microstructures on experimentally relevant length and time scales so as to accelerate and optimize the design of new technological alloys that can directly benefit US industries.
Task Progress:	To this day, the kinetics behavior of side-branches and the development of the grain envelope shape remain partially understood. We have used the Dendritic Needle Network (DNN) model developed in the previous project to investigate the growth behavior and the competition of dendritic side-branches under isothermal condition. Based on DNN simulations, three growth stages have been identified and analytical growth laws have been obtained for each of them. This allows to predict an analytical grain envelope that shows reasonable agreement with the one extracted from the DNN simulations. In its original formulation, the DNN model is only able to investigate the growth of grains with a single orientation. This limitation is due to the fact that dendrites are oriented along rows and columns of the cubic finite difference grid used to integrate the solute field. The extension of the DNN model to various crystalline orientations has been performed thanks to the introduction of a new field marking the grid points belonging to the solid. We have checked that this new formalism reproduces the expected stationary velocity of free dendrites of various orientations. In addition, the nucleation of new grains and the primary growth stage of spherical growth of the nuclei have also been incorporated into the same framework. This model then includes all the essential ingredients to model accurately the columnar-to-equiaxed transition. This transition, of significant importance in industrial solidification processes, is due to the nucleation of equiaxed grains ahead of the columnar front which block the progression of the columnar dendrites. Preliminary simulations have been performed and show promising results. The third part of this project consists of investigating the competitive growth of columnar grains with different orientations in a temperature gradient. The mechanisms involved in two-dimensional grain competition have been investigated in the framework of the previous project. This study sheds light on the mechanisms responsible for grain elimination in 2D. However, an accurate description of the experimental findings requires a full three-dimensional model. We have developed a 3D phase-field model able to model accurately the growth of misoriented grains with 2 misorientation angles. The model has been validated and a set of simulations is currently running to investigate the three-dimensional dynamics of misoriented grain competition We are also considering using the new version of the DNN to complete the results of the phase-field simulations, computationally expensive in 3D.
Bibliography Type:	Description: (Last Updated: 05/10/2026)
Abstracts for Journals and Proceedings	Karma A. "Multiscale modeling of dendritic microstructures: bridging the dendrite tip and grain scales." ICASP-4: 4th International Conference on Advances in Solidification Processes, Old Windsor, UK, July 8-11, 2014. ICASP-4: 4th International Conference on Advances in Solidification Processes, Old Windsor, UK, July 8-11, 2014. Abstract Book. , Jul-2014
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Articles in Peer-reviewed Journals	Tourret D, Karma A. "Growth competition of columnar dendritic grains: A phase-field study." Acta Materialia. 2014, in press. (as of September 2014). , Sep-2014

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